

R Data Structures Overview

Data Structures in *R*

Data structures are used to store and organise values.

R provides many built-in data structure. Each is used to handle data in different ways:

- Vectors
- Lists
- Matrices
- Arrays
- Data Frames

Vectors

A vector is the most basic data structure in *R*. It contains a list of items of the same type.

Example

```
# Vector of strings
fruits <- c("banana", "apple", "orange")
fruits
# Output: [1] "banana" "apple" "orange"
```

Lists

A list can hold different types of data in one structure. It can combine numbers, strings, vectors, and even other lists.

Example

```
# List of strings
thislist <- list("apple", "banana", 50)
```

```
thislist
# Output:
# [[1]]
# [1] "apple"
#
# [[2]]
# [1] "banana"
#
# [[3]]
# [1] 50
```

Matrices

A matrix is a 2D data structure where all elements are of the same type. It is like a table with rows and columns.

It uses `nrow` and `ncol` to control the size of the matrix.

Example

```
# Create a matrix
thismatrix <- matrix(c(1,2,3,4,5,6), nrow=3, ncol=2)
thismatrix
# Output:
#      [,1] [,2]
# [1,]    1    4
# [2,]    2    5
# [3,]    3    6
```

Arrays

An array is like a matrix but can have more than two dimensions. It stores elements of the same type in multiple dimensions.

Example

```

# An array with one dimension with values ranging from 1 to 12
thisarray <- c(1:12)
thisarray
# Output: [1]  1  2  3  4  5  6  7  8  9 10 11 12

# An array with more than one dimension
multiarray <- array(thisarray, dim = c(2, 3, 2))
multiarray
# Output:
# , , 1
#
#      [,1] [,2] [,3]
# [1,]    1    3    5
# [2,]    2    4    6
#
# , , 2
#
#      [,1] [,2] [,3]
# [1,]    7    9   11
# [2,]    8   10   12

```

Data Frames

A data frame is like a table in a spreadsheet. It can hold different types of data across multiple columns.

Example

```

# Create a data frame
DataFrame <- data.frame (
  Training = c("Strength", "Stamina", "Other"),
  Pulse = c(100, 150, 120),
  Duration = c(60, 30, 45)
)

```

```
# Print the data frame
DataFrame
# Output:
#   Training Pulse Duration
# 1 Strength    100        60
# 2  Stamina    150        30
# 3   Other    120        45
```

Summary

Data Structure	Contains	Same Type?	Use Case
Vector	Single row of values	Yes	Simple sequences
List	Multiple types of values	No	Grouped mixed data
Matrix	2D same-type values	Yes	Tables with numeric data
Array	Multi-dimensional values	Yes	3D or higher-dimensional data
Data Frame	Columns of mixed types of values	No	Tabular data